Fine-Grained Recognition for Oriented Ship Against Complex Scenes in Optical Remote Sensing Images

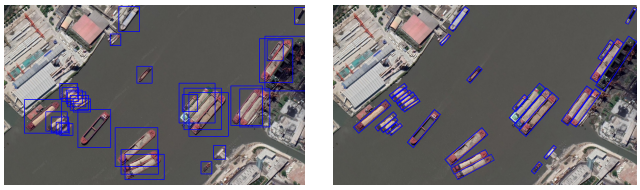
Yaqi Han, Xinyi Yang, Tian Pu, and Zhenming Peng, *Member, IEEE*

***Abstract*— Remote sensing ship recognition is widely used in civil and military applications, such as national defense construction, fishery management, and navigation supervision. Unfortunately, limited by the lack of public datasets for fine-grained ship recognition, current studies mainly focus on ship detection or coarse-grained ship recognition, whereas fine-grained recognition task is left out. The main challenges for fine-grained ship recognition include: 1) complex scenes and 2) ship characteristics of arbitrary orientation, dense distribution, and huge scale and appearance variation. Aiming at the challenges above, we propose a novel efficient information reuse network (EIRNet) and establish a public 20-class Dataset for Oriented Ship Recognition (DOSR). In our EIRNet, considering recognition robustness to multiscale ships, a dense feature fusion network (DFF-Net) with two fusion directions is designed to maximize the utilization of multilayer information and reduce information redundancy. Then, the fused feature maps are refined by a dual-mask attention module (DMAM) to improve performance in dense and clutter scenes by enhancing the distinction between ships and suppressing clutter. Furthermore, Mask-RPN improves the efficiency of generating proposals by reusing the attention mask. Finally, we introduce a concept of upper level class to mine interclass relationships, which further improves the recognition accuracy. Extensive experiments demonstrate that our EIRNet achieves state-of-the-art performance on DOSR and another popular public dataset called HRSC2016.**

***Index Terms*— Attention mechanism, feature fusion networks, fine-grained ship dataset, information reuse, oriented ship recognition.**

I. Introduction

SHIP-BASED activities play a key role in a variety of marine activities. Ship recognition is an image processing technology that deals with detection and classifies ship instances in digital images [1]. With the rapid development of remote sensing fields, remote sensing ship recognition is widely used in more and more marine applications, such as national defense construction, fishery management, navigation supervision, and marine rescue. Compared with other remote sensing data sources, optical remote sensing images are easy to understand intuitively, contain abundant semantic information, and have higher resolution, which brings advantages to the accurate recognition of ship targets. Due to the quality improvement of optical remote sensing images, ship recognition in optical remote sensing images, especially in complex backgrounds, aroused much attention in recent years.



(a) (b)

Fig. 1. Different annotation methods. (a) Horizontal bounding boxes. (b) Oriented bounding boxes.

Unfortunately, most studies only focus on ship detection tasks, which only distinguishes ship targets from nonship targets. Some coarse ship recognition approaches only divide ships into rough two to five categories, such as military ship and civilian ship. Little attention is paid to fine-grained ship recognition. In fine-grained ship recognition, ships need to be classified into more detailed ship types, such as tug, cruise, cargo, and tanker. The main obstacle for such a situation is that current public datasets are not detailed enough to support fine-grained recognition tasks. Therefore, there is an urgent need for fine-grained ship recognition datasets.

Compared with general object recognition tasks, there are three main difficulties in fine-grained ship recognition in remote sensing images. First, the remote sensing images have inherent characteristics of low quality due to their low resolution and severe noise interference. Second, there are various complex scenes in the ship recognition task. Some typical scenes include small scenes, dense scenes, clutter scenes, and huge scale variations scenes. Moreover, these scenes often do not appear alone. For example, ships in some images can be both small and dense. Third, ship targets have the characteristics of arbitrary orientation, excessive aspect ratio, dense distribution, and huge scale and appearance variations. These unique characteristics present the greatest challenges to fine-grained ship recognition tasks.

1) *Arbitrary Orientation:* From the bird-eye perspective of remote sensing images, ships often appear in arbitrary orientations, which means that traditional horizontal bounding boxes are no longer applicable. As displayed

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The authors are with the School of Information and Communication Engineering and the Laboratory of Imaging Detection and Intelligent Perception, University of Electronic Science and Technology of China, Chengdu 610054, China (e-mail: hanyaqi@std.uestc.edu.cn; xyyang_2015@163.com; putian@uestc.edu.cn; zmpeng@uestc.edu.cn).

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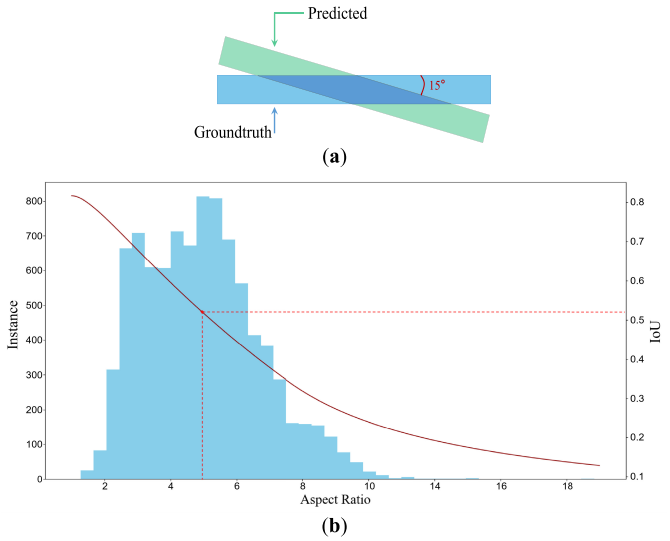

Fig. 2. Aspect ratio characteristic of ships. (a) Example of IoU between high aspect ratio oriented boxes with a slight angle difference of 15°. (b) Blue histogram refers to the distribution of aspect ratio in a merged dataset (our DOSR and HRSC2016 [2]). The dark-red curve displays the aspect ratio versus IoU when the angle between boxes differs by 15°, and the red dotted line represents the average aspect ratio of 4.93 and its corresponding IoU of 0.52.

in Fig. 1(a), horizontal bounding boxes will lead to unacceptable redundant overlap between objects. Therefore, it is necessary to use oriented bounding boxes shown in Fig. 1(b).

2) *Excessive Aspect Ratio:* As shown in Fig. 2(a), in the case of high aspect ratio, a small angle deviation between the predicted box and ground-truth box will cause a rapid decrease in Intersection over Union (IoU), which is the basis of the evaluation index and sensitive hyperparameters in nonmaximum suppression (NMS). Unfortunately, the statistics in Fig. 2(b) indicate that the IoU of more than 50% of instances falls below 0.52 with a 15° angle difference.
3) *Dense Distribution:* In dense scenes, multiple side-by-side ships are easily recognized as one ship. Besides, one predicted box may straddle between two dense ships.
4) *Huge Appearance and Scale Variations:* In fine-grained ship recognition, some ship categories have huge intraclass appearance differences due to different regulations in different countries or regions. Some ship classes with similar usages also have interclass similarities, which may confuse the network's classification branches. Besides, scale variation can be extremely huge, causing a high risk of missing or wrong recognition, especially for tiny ships.

In decades, many efficient approaches have been proposed to solve ship recognition tasks in aerial images. There are two main development branches for these approaches: feature-design- and deep-learning-based methods. Feature-design-based models mainly adopt a "coarse-to-fine" two-stage strategy. In the first stage, the sea area is segmented as Region of Interest (RoI) to exclude false alarms in land area; then, candidate regions are proposed for the subsequent recognition process. The main algorithms include algorithms based on statistical data [3]–[6], visual saliency [7], [8], frequency analysis [9]–[12], and reflection difference [13]. The second stage aims to extract features and classify the candidate regions with classifiers, such as SVM and Adaboost. Commonly handcraft features used in this stage include LBP features, Harr features, HOG features, and their improved models. Feature-design-based models perform well in applications with simple scenes, such as infrared small target detection [14]–[16]. However, the requirement of complete prior knowledge makes the recognition accuracy highly dependent on the design of handcraft features, incurring poor practicability in complex scenarios. The other approach branch is deep-learning-based models that have achieved great success in computer vision applications. Feature extractors and classifiers in such approaches are obtained through a large amount of data-driven training instead of relying on manual prior design, making it more robust in different scenarios. There are two typical strategies for deep-learning-based algorithms: two-stage networks represented by Faster-RCNN [17] and single-stage networks represented by YOLO [18] and SSD [19]. Faster-RCNN utilizes a region proposal network (RPN) to generate proposals and then obtains final recognition results based on the proposals through classification and regression branches. YOLO and SSD directly complete classification and bounding box regression work at one time based on feature maps extracted by backbone networks. Two-stage networks have high accuracy but lack speed, whereas single-stage networks are faster but sacrifice accuracy to some extent. Although networks mentioned above perform well in general datasets, such as ImageNet and MS COCO, as we analyzed above, the horizontal bounding box of Faster-RCNN, YOLO, and SSD is not suitable for ship recognition tasks. In addition, their performance is challenged by problems, such as clutter, dense distribution, and excessive aspect ratio.

To improve the accuracy and robustness of fine-grained ship recognition in complex scenes, we propose a novel efficient information reuse network (EIRNet), which is an end-to-end network with a multitask training strategy. In view of the accuracy superiority of two-stage networks, our EIRNet adopts a two-stage structure, combining with novel feature fusion networks and attention mechanisms. The components in EIRNet are well-designed in response to the challenges mentioned above. Our EIRNet achieves state-of-the-art performance both on DOSR and HRSC2016. The main contributions of this article are given as follows.

1) To maximize utilization of multilayer information, we develop a dense feature fusion network (DFF-Net) to efficiently fuse information from all layers by combining top-down and bottom-up fusion strategies with no extra memory space, which improves the recognition ability for multiscale targets.
2) For dense distribution and complex background, we proposed a supervised pixel-level dual-mask attention module (DMAM), which can not only suppress noise and highlight targets but also effectively distinguish between closely docked ships. The mask generated by DMAM is reused in the proposed Mask-RPN to further boom the network performance.

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3) For the requirement of oriented bounding boxes and accurate fine-grained recognition, a concept of upper level class is introduced to mine interclass relationships and reduce the risk of overfitting. Besides, the smooth L1 loss is modified to solve loss mutation and box straddle problems.
4) We established a new 20-class public dataset called DOSR for fine-grained ship recognition tasks in remote sensing images.

The rest of this article is organized as follows. Section II briefly reviews the related work on ship recognition approaches based on deep learning and current publicly available datasets for the ship recognition task. Section III introduces the details of the proposed methodology. Section IV displays the information of the new dataset and discusses the experiment results of EIRNet. Section V concludes the works of this article.

II. Related Work

A. Ship Recognition Based on Deep Learning

Since the rapid rise of deep learning [20]–[23], deep-learning-based object recognition approaches for remote sensing images have constantly developed in the last decade, which shows great superiority to traditional feature-design-based models. Unfortunately, affected by the lack of datasets, studies on ship recognition remain rare. Therefore, in this section, some networks designed for general remote sensing images, which can be applied to ship recognition, will also be included.

To address the challenges mentioned in Section I, researchers have made various efforts. Different from general object detection, fine-grained ship recognition is required to not only locate ships precisely but also classify ships accurately against huge appearance variations. Zhang *et al.* [24] designed an additional backbone to extract attribute information, such as aspect ratio and scale, and utilized SVM to make the final classification based on concated attribute features and multiscale image features. Feng *et al.* [25] first located ships and then cut out the ship patch to further classify fine-grained categories. Sun *et al.* [26] employed a cascade R-CNN structure with different NMS thresholds in each stage to realize coarse-to-fine recognition. These strategies all prolong the recognition process and increase the computational burden.

There are two main approaches to deal with the demand for oriented bounding boxes: 1) angle-based regression adopted by Feng *et al.* [25], Ma *et al.* [29], and Zhang *et al.* [30] based on rotational region CNN (R2CNN) [30], which introduces angle parameter to regression branch and 2) point-based regression applied in [32], [33], and [71] directly predicts offsets or coordinates of four vertexes of oriented bounding boxes. Based on Xu *et al.* [32], Huang *et al.* [33] constrained the box shape by utilizing a diagonal length ratio to regress bounding boxes more accurately. To further reduce the adverse effect of orientation on recognition accuracy, Yu *et al.* [34] first utilized angle-based regression to coarsely locate ships and then aligned the boxes through point-based regression. Aiming at the problem CNN features that are not equivariant to the rotation, Han *et al.* [35] proposed a rotation-equivariant CNN (ReCNN) to extract rotation-equivariant features, which efficiently improved the prediction accuracy of orientation.

Fang *et al.* [36] found that the detection performance for objects with a high aspect ratio was much lower than the overall average. The mainstream strategy to deal with the excessive aspect ratio of ships is still increasing the range of aspect ratio of anchors. The cluster-based approach proposed in [36] is an effective attempt although it is not end-to-end.

Aiming at improving recognition performance against complex scenes, such as severe clutter and dense distribution, the feature fusion and attention mechanism has been widely utilized and developed in ship recognition to enhance the feature representation capability of networks. For example, Jiao *et al.* [59] constructed a densely connected multiscale neural network (DCMSNN) based on feature pyramid networks (FPNs) [57] to avoid the inference of inshore background. Feng *et al.* [25] and Guo *et al.* [37] employed dilated convolution and deformable convolution, respectively, to strengthen multiscale semantic information in the fusion process. There are also some new networks that fuse features by constructing new backbones. Dual-tunnel CNN [38] analyzes the joint features from the fused information spatially and spectrally. Zhang *et al.* [39] design a PToP CNN to cover a convolutional filter bank with three tunnels. Unfortunately, most feature fusion structures only considered either top-down or bottom-up fusion direction, whereas another information flow was ignored. A novel feature fusion called MSR-FPN [46] achieved high performance by merging the top-down and bottom-up branches through a dilated convolutional parallel structure. Anchor-free CPS-Det [40] combined NAS-FPN [61] and FCOS [41] to search for the best fusion pattern and improve dense box detection performance. CSA-DRF [33] improved the bidirection fusion through simplifying SConvs and constructing a dynamic receptive field mechanism. However, most current bidirection fusion networks construct an extra bottom-up fusion pipeline, which costs more memory space. For attention mechanisms, many ship detection approaches [24], [42], [43] simply embedded popular attention mechanisms in other fields, such as CBAM [58] and RNN-based attention module [44], into their frameworks. Little work designed attention mechanisms aiming at specific characteristics of ships. Fortunately, some efficient attention mechanisms have emerged in the general remote sensing recognition field. Yang *et al.* [45] proposed a network called SCRDet and proved the effectiveness of the supervised attention mechanism for clutter suppression and target enhancement. Huang *et al.* [46] designed the nonlocal-aware pyramid attention to solve the misalignment between the network's emphasis and the real objects in the images with high performance.

B. Current Public Datasets

Datasets play a significant role in data-driven methodologies. In recent years, the public availability of more and more remote sensing datasets has promoted the development of research on object detection and recognition in remote

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TABLE I
CURRENT PUBLIC DATASETS CONTAINING SHIP TARGETS

Datasets	Image amount	Instance amount	Category amount	Main categories	Bounding box	Classification level	Year
BCCT200	800	800	4	barge, tanker, etc.	oriented	L2	2011
NWPU VHR -10	800	3775	10	ship, airplane, etc.	axis-aligned	L1	2014
HRSC2016	1061	2886	1/4/19	ship	oriented	L1, L2, L3	2016
DOTA	2806	188282	15	ship, airplane, etc.	oriented	L1	2017
HRRSD	21,761	55,740	13	ship, airplane, etc.	axis-aligned	L1	2017
Airbus Ship Detection	208162	56031	1	ship	oriented	L1	2018
xView	1128	~1,000,000	60	ship, airplane, etc.	axis-aligned	L1	2018
DIOR	23463	192472	20	ship, airplane, etc.	axis-aligned	L1	2018

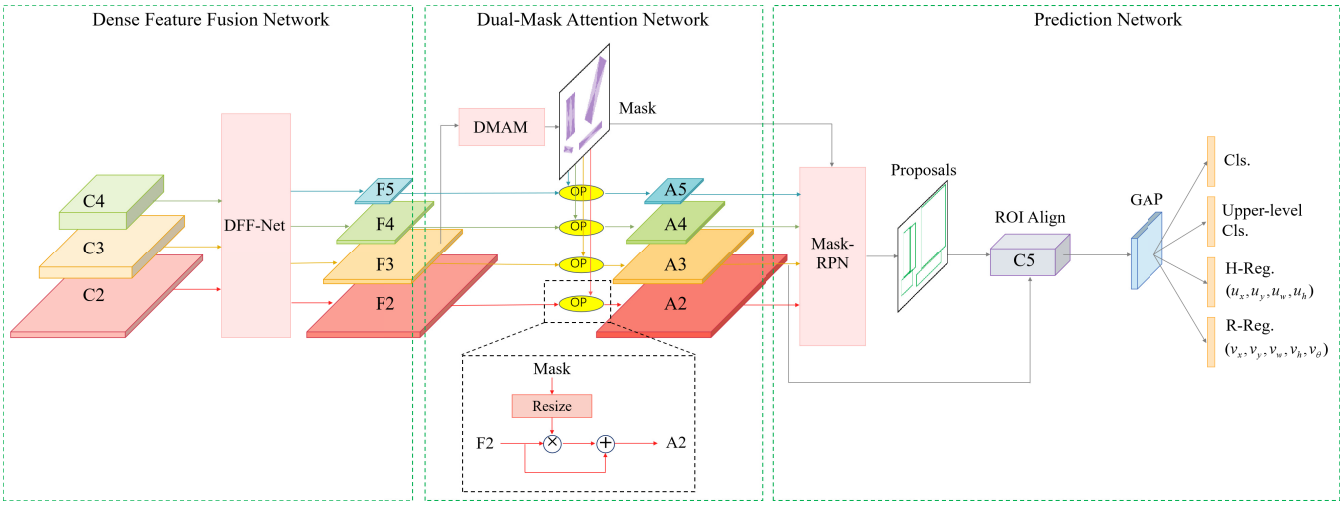

Fig. 3. Architecture of the proposed EIRNet. EIRNet can be divided into three subnetworks according to their functions: 1) DFF-Net efficiently fuses information from all layers; 2) dual-mask attention network suppresses clutter and highlight targets information through DMAM; and 3) prediction network utilizes Mask-RPN, C5 block, global average pool (GAP), and four prediction branches to predict the class and location of bounding boxes.

sensing images. However, compared with the vigorous emergence of general image datasets, the amount and diversity of remote sensing image datasets are not abundant enough, and datasets for ship detection and recognition are rather rare, especially for fine-grained ship recognition. Table I lists the information of current public datasets containing ship targets [2], [47]–[53]. Among the eight datasets, six of them are just applicable to level-1 task: distinguish between ships and nonship targets. Only two datasets, BCCT200 and HRSC2016, classify ships into more detailed types. Ships in BCCT200 are roughly classified into four level-2 categories: barge, tanker, cargo, and container. HRSC2016 classifies ships at three different levels simultaneously. More specifically, the level-2 classification includes five categories, such as warcraft and merchant ship, and the level-3 classification contains 19 categories, such as car carrier and medical ship. As the only fine-grained dataset at present, HRSC2016 is used as a benchmark in many ship recognition studies. Unfortunately, for the level-3 classification of HRSC2016, there are six categories with no more than ten ship instances, and the number of instances in 11 categories is less than 50. Moreover, some nonnegligible small ships are not annotated in HRSC2016, which reduces the credibility of HRSC2016 as a benchmark to testify networks. In this article, in order to break the limitation of lacking datasets for fine-grained ship recognition, a new 20-class level-3 Dataset for Oriented Ship Recognition (DOSR) is established. The detailed information of DOSR is introduced in Section IV.

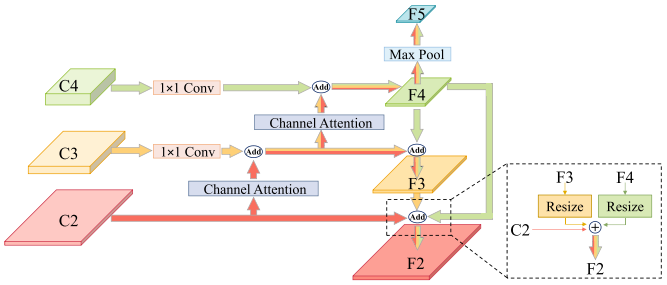

Fig. 4. Structure of DFF-Net. The colors of the arrow imply from which layer that it contains information.

III. METHODOLOGY

The overall architecture of the proposed EIRNet is illustrated in Fig. 3. EIRNet is an end-to-end two-stage network with multitask training strategy. In the first stage, feature maps are extracted and densely fused by DFF-Net, decluttered and refined by DMAM, and input to the Mask-RPN to generate proposals. In the second stage, each proposal is processed by four branches jointly to predict class scores and bounding boxes. Final prediction results are obtained after the post-processing of R-NMS [29].

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A. Dense Feature Fusion Network

How to extract features from input images efficiently is a crucial problem for object detection and recognition tasks. In recent years, many researchers have made explorations in the feature extraction process and proposed many excellent backbone networks [23], [54]–[56]. There is one core idea in common for these backbones, that is, as the layers go higher, the receptive fields continuously expand through pooling, dilated convolution, or other approaches. Thus, higher layer feature maps have rich semantic information but poor location information, whereas lower layer feature maps have high resolution but lack semantic information. Therefore, it is necessary to fuse features from different layers to make a trade-off between semantic information and location information. However, many popular top-down fusion networks [57], [59] only adopt top-down fusion structure, whereas bottom-up fusion is ignored. Although some studies [33], [46], [60]–[62] also consider the bottom-up fusion direction, most of them add an extra fusion pipeline, which sacrifices memory space and makes the network structure redundant and cumbersome. Therefore, we design a DFF-Net to introduce a bottom-up fusion direction without adding additional memory burden. Fig. 4 depicts the structure of the proposed DFF-Net.

Feature maps first unify the channel dimension through 1×1 convolution. In the fusion process, we take into account both top-down and bottom-up fusion directions. For the bottom-up part, the lower layer feature map is input to the channel attention module to adjust the weight of each channel before being merged upward and then is elementwisely added to the higher layer feature map. Due to the different resolutions of feature maps, the addition operation is always accompanied by the resize operation, which is implemented with the bilinear interpolation algorithm. For the top-down part, different from general FPN, which only merges two adjacent layers, our DFF-Net merges all layers above the current layer. Since C_5 block is applied to the head of the prediction network, there are only three input layers. In order to adapt to a wider range of anchor scales, we perform a max-pooling operation on F_4 to obtain F_5. The intuitive expression of the above process is given as follows:

$$F_4 = \text{Conv}_{1\times1}(C_4) + \text{UpSample}\left\{\text{CA}\left[\text{Conv}_{1\times1}\ (C_3) + \text{UpSample}\ (\text{CA}\ (C_2))\right]\right\} \quad (1)$$

$$F_3 = \text{DownSample}\ (F_4) + \text{Conv}_{1\times1}\ (C_3) + \text{UpSample}\ [\text{CA}\ (C_2)] \quad (2)$$

$$F_2 = \text{DownSample}\ (F_4) + \text{DownSample}\ (F_3) + C_2 \quad (3)$$

$$F_5 = \text{MaxPool}\ (F_4) \quad (4)$$

where $C_i (i = 2, 3, 4)$ is feature map of layer i and $F_i (i = 2, 3, 4, 5)$ is the feature map fused by DFF-Net. MaxPool means the stride $= 2$ max pooling. CA represents the channel attention module, which adjusts the channel weights according to their contribution to the final recognition task. In this article, we employ the channel attention module in CBAM [58] as our CA. The structure is shown in Fig. 5, and the parameter settings are consistent with [58]. The channel attention module highlights the salient information in rich but redundant features and enhances the representation ability of the network by smashing the tradition of simply adding up feature maps from different layers equally.

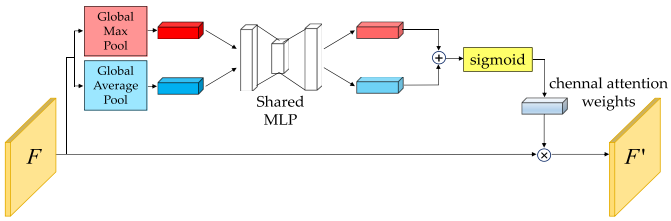

Fig. 5. Structure of the channel attention module in CBAM.

In DFF-Net, every output feature map combines the information from each backbone layer, maximizing the representation ability of the fused information by an efficient information reuse approach. The full information integration improves the accuracy of recognizing multiscale objects, especially small objects. Moreover, the recognition of ship types always depends on some detailed characteristics of ships, such as deck facilities and bow curvature. The bottom-up fusion direction will strengthen the detailed semantic information of each layer, which is beneficial to accurate fine-grained recognition.

B. Dual-Mask Attention Module

Complex and dense scenes are one of the main culprits for the poor performance of remote sensing ship recognition tasks. For example, as shown in Fig. 6(b), ships docked side by side are difficult to distinguish in feature maps because of their blurred boundaries. Due to the complexity of the background, some excessive clutter can interfere with or even overwhelm the object information. These problems will exert an adverse impact on the final recognition performance. Many attention mechanisms [58], [63]–[66] have been proposed to solve these problems. However, most approaches are unsupervised. Thus, they cannot guide the network to learn for a specific purpose. An inspiring supervised network proposed in [45] did good work to suppress background clutter and strengthen the object information.

Inspired by Yang *et al.* [45], we propose a supervised pixel-level dual-mask attention network, and the core design is DMAM. Since channel attention has been discussed in DFF-Net, DMAM mainly focuses on spatial attention. Fig. 7 illustrates the framework of the dual-mask attention network. Mask-1 and Mask-2 are binary images generated from the ground truth. Values of object area are set to 1, and values of background area are set to 0. The shape of Mask-2 is designed based on the fact that the key differences between similar ships, such as type numbers and deck patterns, often lie in the head and tail of the ship. The shortcut structure ensures that the introduction of Mask 2 will not lead to the loss of central information. The contour of Mask-1 and Mask-2 can be formulated as

$$\text{Contour}_{\text{Mask}-1} = (x_1, y_1, x_2, y_2, x_3, y_3, x_4, y_4) \quad (5)$$

$$\text{Contour}_{\text{Mask}-2} = (x_1, y_1, x_3, y_3, x_2, y_2, x_4, y_4) \quad (6)$$

where $(x_i, y_i)(i = 1, 2, 3, 4)$ is clockwise vertex coordinates of the ground truth.

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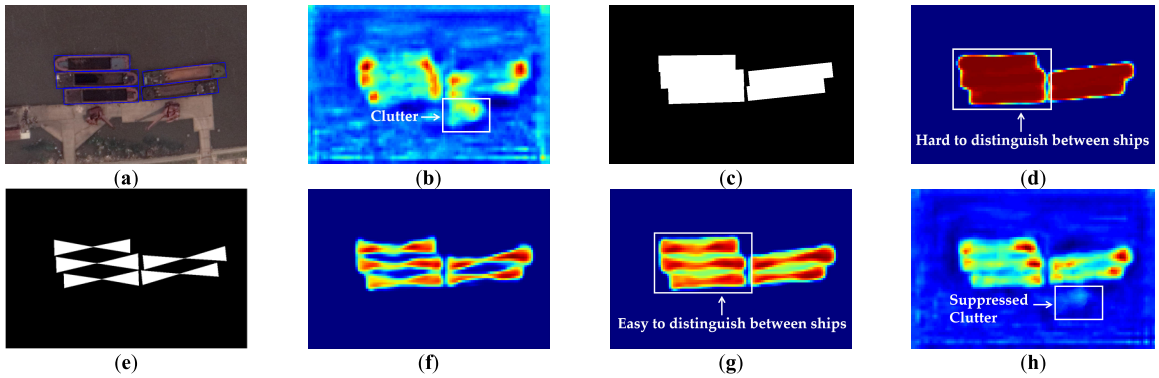

Fig. 6. Example of feature maps and masks in dense scenes. (a) Ground truth. (b) Original feature map. (c) Mask-1 generated from the ground truth. (d) M_1 produced by DMAM under the supervision of Mask-1. (e) Mask-2 generated from the ground truth. (f) M_2 produced by DMAM under the supervision of Mask2. (g) Final mask M. (h) Feature map after the operation with mask M.

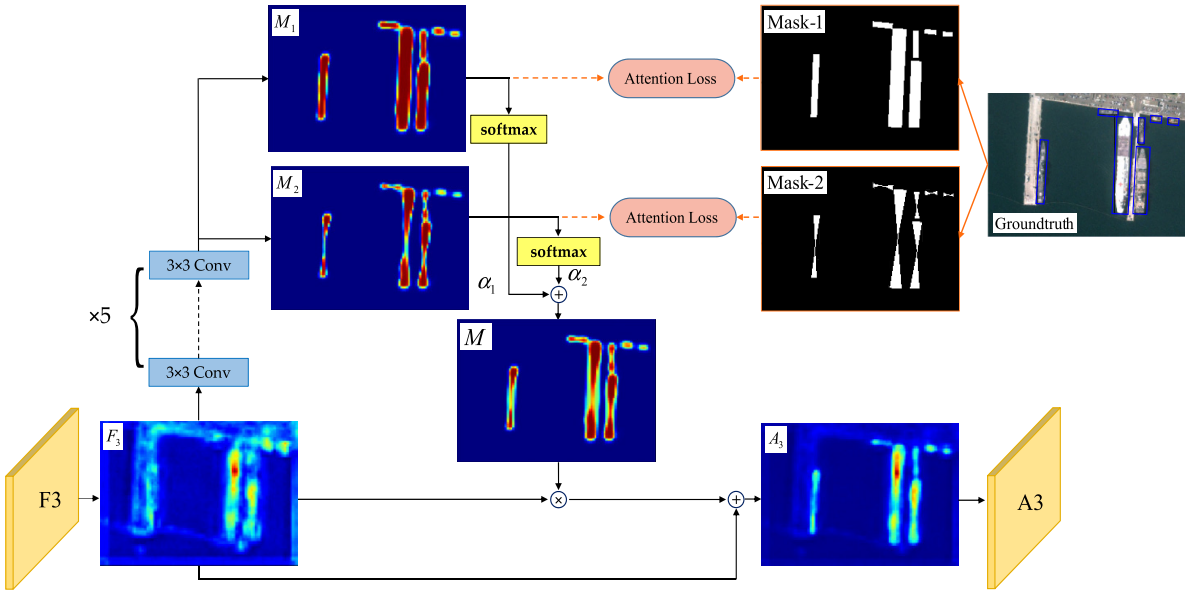

Fig. 7. Framework of the dual-mask attention network.

With five 3×3 convolutional layers, Mask-1 and Mask-2 guide the network to learn the corresponding two masks: M_1 and M_2. After the SoftMax operation, the values of M_1 and M_2 are in the range of [0, 1]; then, they are added with weights to obtain the final pixel-level attention mask M. Inputs of dual-mask attention network are feature maps F_i $(i = 2, 3, 4, 5)$ from DFF-Net, and the output of the network A_i $(i = 2, 3, 4, 5)$ is

$$A_i = F_i + F_i \times (\alpha_1 M_1 + \alpha_2 M_2) \tag{7}$$

where hyperparameter α_1 and α_2 are weight coefficients. In practice, to reduce unnecessary computations, we use F_3 to calculate mask M through DMAM, and then, the obtained M is resized and shared to all layers (as shown in Fig. 3).

It is worth noting that, if we only use the mask shown in Fig. 6(c) to train the network, M_1 produced by DMAM [see Fig. 6(d)] can only make a distinction between targets and background but also it is hard to distinguish between targets and targets. Although the response of the nontarget area is well suppressed, dense objects are highlighted at the same time, and the boundary ambiguity problem shown in Fig. 6(b) remains unsolved. Thus, Mask-2 [see Fig. 6(e)] is introduced to guide the learning process jointly with Mask-1. We can observe that the final mask M [see Fig. 6(g)] not only magnifies the distinction between targets and background but also makes a great distinction between targets and targets. Fig. 6(b) and (h) shows the feature maps before and after operating with mask M. It is obvious that the background clutter in the original feature map is well suppressed, and the boundary of the dense target becomes much clearer.

C. Prediction Network

1) Mask-RPN: Since the receptive field is affected by the resolution of the feature maps, higher level feature maps with stronger semantic information are adapted to larger scale target recognition, whereas the lower level feature maps with stronger position information are suitable for smaller scale target recognition. Therefore, for the multiscale recognition task, we employ an improved RPN called Mask-RPN to generate proposals from multilayer feature maps with different

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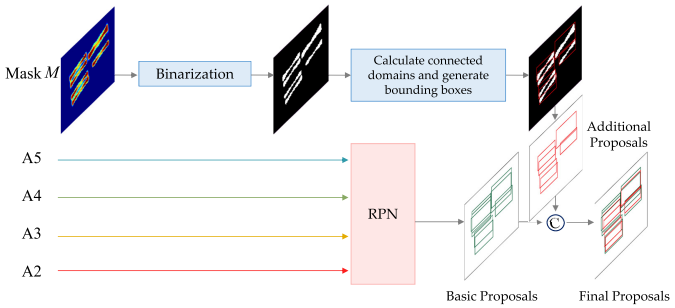

Fig. 8. Structure of Mask-RPN. In Mask-RPN, basic proposals generated by traditional RPN and additional proposals calculated from mask M are concated to get the final proposals.

anchor scales. The structure of the Mask-RPN is illustrated in Fig. 8.

The final proposals generated by Mask-RPN have two components: 1) basic proposals generated by traditional RPN [17] and 2) additional proposals calculated from mask M. For basic proposals, we use horizontal anchors in RPN because angle parameters introduced in rotated anchors will multiply the number of anchors [29], [30], and the accuracy gains do not match the additional computational expenses. We assign four scales $\{64, 128, 256, 512\}$ anchors to $\{A_2, A_3, A_4, A_5\}$, respectively. Three aspect ratios {1:1, 1:2, 2:1, 1:3, 3:1, 1:4, 4:1} are adopted for anchors of each scale. The NMS threshold is set to 0.7. In the training process, the numbers of top-scoring boxes to keep before and after applying NMS are 12 000 and 2000, and in the testing process, we obtain 600 proposals after applying NMS to 6000 top scoring boxes. Other hyperparameters, such as the threshold of positive and negative samples, are consistent with [17].

Based on the principle of maximizing the reuse of existing information, we introduce additional proposals to improve the performance of RPN. Through extensive observation, we found that mask M calculated by DMAM can match the ground truth well in the early training stage. Masks learned by DMAM under different iterations in Fig. 9 validate this opinion. Therefore, we exploit mask M to generate additional proposals. This process only needs two steps. First, apply threshold segmentation operation to the single-channel mask M to get a binary image. Second, compute connected domains in the binary image, and output their minimum bounding boxed as the additional proposals. Basic proposals and additional proposals are concated to get the final proposals. Final proposals are mapped to A_3 feature map and input to prediction network to complete subsequent regression and classification. For the choice of head of the prediction network, we substitute two fully connected layers generally used in many algorithms [17]–[23] with the C_5 block of the backbone and global average pooling (GAP), which reduces network parameters effectively. Besides, the pretrained C_5 module makes the training process converge faster, and GAP greatly decreases the risk of overfitting under class imbalance.

2) Prediction Branches:

a) Branches for bounding box regression: Compared with general multiclass detection networks, one of the new branches in EIRNet is the oriented bounding box regression branch. Although this new branch can meet the requirement of predicting rotated bounding boxes, we retain the traditional horizontal bounding box regression branch since the horizontal bounding box regression branch still makes an indispensable contribution to network accuracy. Relevant experiments and analyses are shown in Section IV.

We employ a five-parameter tuple $(x,\ y,\ w,\ h,\ \theta)$ to describe the ship's position. As shown in Fig. 10, (x, y) demotes the center point coordinate of the rotated bounding box. w and h denote the length of the shorter side and the longer side, respectively. θ is the clockwise angle between the bounding box and reference rectangle. The regression of oriented bounding boxes is given as follows:

$$\begin{aligned} v_x &= (x - x_p)/w_p, \quad v_y = (y - y_p)/h_p \\ v_w &= \log(w/w_p), \quad v_h = \log(h/h_p), \quad t_\theta = \theta - \theta_p \\ v'_x &= (x' - x_p)/w_p, \quad v'_y = (y' - y_p)/h_p \\ v'_w &= \log(w'/w_p), \quad v'_h = \log(h'/h_p), \quad t'_\theta = \theta' - \theta_p \end{aligned} \tag{8}$$

where x, y, w, h, and θ represent the center point coordinates, width, length, and angle of the rotating bounding box, respectively. Variables x, x_p, and x' are for the anchor box, predicted box, and ground-truth box, respectively (likewise for y, w, h, and θ).

In the horizontal box regression branch, we obtain ground-truth horizontal bounding boxes by calculating minimum bounding rectangles of oriented bounding boxes. Four parameters $(x,\ y,\ w,$ and $h)$ are exploited to describe the ship position, where (x, y) are the coordinates of the center point of the horizontal bounding box, and w and h are the length and width of the horizontal box. The regression of the horizontal bounding box is consistent with [17].

b) Branches for classification prediction: As mentioned in Section II, current fine-grained ship recognition networks either directly predict all ship categories, such as general object recognition, or introduce additional feature extraction structures with heavy calculation and memory burden. In fact, fine-grained ship categories are not absolutely irrelevant. Instead, they share some common characteristics in upper level classes. To mine the potential interclass relationships among so many fine-grained ship categories and alleviate the negative impact of class imbalance, we propose a new upper level classification branch to assist the fine-grained classification branch with little calculation cost. Fig. 11 describes the concept of upper level class, where L2 is the upper level class of L3, and L1 is the up-level class of L2. In the training stage, the fine-grained classification branch and the upper level classification branch predict the L3 class and the L2 class, respectively. In the testing stage, only the fine-grained classification branch works to output the scores of L3 classes. Therefore, the introduction of upper level classification branch costs no extra computation expense in the testing stage with little additional memory space.

Ablation experiments show that this simple auxiliary branch improves the accuracy of fine-grained classification efficiently. Apart from the collocation of L3+L2, we also try to utilize all three classification branches of L3+L2+L1 to conduct prediction. Nevertheless, the prediction results are not as good

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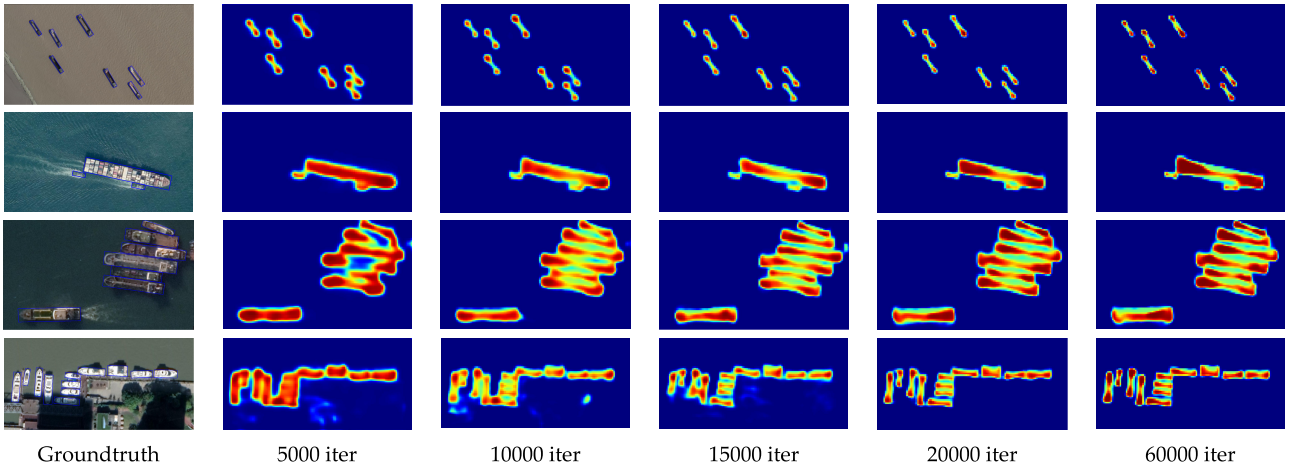

Fig. 9. Visualization results of mask M calculated by DMAM at different training iterations. The first column is ground truth; the second to sixth columns are mask M calculated at 5000, 10 000, 15 000, 20 000, and 60 000 training iterations, respectively. Mask M calculated at a very early stage (10 000 iterations) is already passable to match the ground truth, and the results can completely meet satisfaction after 20 000 iterations.

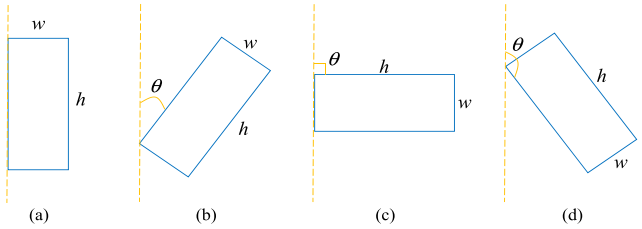

Fig. 10. Parameters of the rotated bounding box. (a) Reference rectangle, $\theta = 0$. (b) $\theta \in (0, \pi/2)$. (c) $\theta = \pi/2$. (d) $\theta \in (\pi/2, \pi)$.

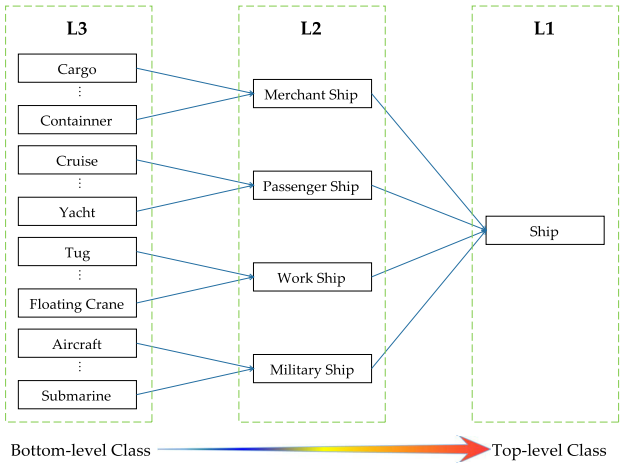

Fig. 11. Concept of the upper level class. L3 is the fine-grained ship type, L2 is the ship category, and L1 is the ship class. L2 is the upper level class of L3, and L1 is the up-level class of L2.

as that of the combination of L3+L2. See Section IV for a detailed discussion.

D. Loss Function

We adopt a multitask loss function for end-to-end training. There are five learning tasks totally, including rotated box regression, horizontal box regression, pixel-level attention mask generation, fine-grained classification, and upper level classification. The loss function is given as follows:

$$
\begin{aligned}
L = {} & \frac{\lambda_1}{N}\sum_{n=1}^{N} t'_n \sum_{j\in\{x,y,w,h,\theta\}} \frac{L_{\text{reg}}\left(v'_{nj}, v_{nj}\right)}{\left|L_{\text{reg}}\left(v'_{nj}, v_{nj}\right)\right|}\left|\text{Factor_IOU}\right| \\
& + \frac{\lambda_2}{N}\sum_{n=1}^{N} t'_n \sum_{j\in\{x,y,w,h\}} L_{\text{reg}}\left(u'_{nj}, u_{nj}\right) \\
& + \frac{\lambda_3}{h\times w}\sum_{i}^{h}\sum_{j}^{w}\left[L_{\text{att}}\left(m'^{[1]}_{ij}, m^{[1]}_{ij}\right) + L_{\text{att}}\left(m'^{[2]}_{ij}, m^{[2]}_{ij}\right)\right] \\
& + \frac{\lambda_4}{N}\sum_{n=1}^{N} L_{\text{cls}}(p_n, t_n) + \frac{\lambda_5}{N}\sum_{n=1}^{N} L_{\text{cls_up}}\left(p_n^{\text{up}}, t_n^{\text{up}}\right)
\end{aligned} \tag{9}
$$

$$
\text{Factor_IOU} = -\log(\text{IoU}_n) + \alpha\frac{1}{M-1}\sum_{k=1}^{M-1}\left(e^{|\text{IoU}_{nk}-\text{IoU}_{k_nk}|}-1\right) \tag{10}
$$

where hyperparameters $\lambda_i (i = 1, 2, 3, 4, 5)$ control the trade-off between five tasks, N and M denote the number of predicted boxes and ground-truth boxes, t_n and t_n^{up} are the bottom class labels and upper level class labels, t'_n is a binary value ($t'_n = 0$ for background and $t'_n = 1$ for foreground), p_n and p_n^{up} indicate the probability distribution of bottom classes and upper level classes, v'_{*j} and u'_{*j} represent the predicted oriented bounding box regression vector and horizontal bounding box regression vector, v_{*j} and u_{*j} represent the ground-truth oriented bounding box regression vector and horizontal bounding box regression vector, $m'^{[1]}_{ij}$ and $m^{[1]}_{ij}$ are the predicted and ground-truth pixel values of mask M_1 (likewise for mask M_2), and IoU_n, IoU_{nk}, and IoU_{k_nk} indicate IoU between the predicted box n and its corresponding ground-truth box k_n, IoU between the predicted box n and other ground-truth boxes k, and IoU between grountruth box k_n and ground-truth box k, respectively. $L_{\text{cls_up}}$ and L_{att} are both softmax cross entropy functions, and L_{reg} is smooth L1 functions. L_{cls} is the class-balanced (CB) softmax cross-entropy loss defined in [67], which is efficient for class imbalance problem with

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TABLE II
COMPARISON BETWEEN DOSR AND HRSC2016

Dataset	Image amount	Instance amount	Average instance amount per image	Class amount	Size of training / validation / testing set	Amount of nearshore scenes/ offshore scenes
DOSR	1066	6127	5.75	20	523 / 223 / 320	832 / 234
HRSC2016	1061	2886	2.72	19	436 / 181 / 444	991 / 70

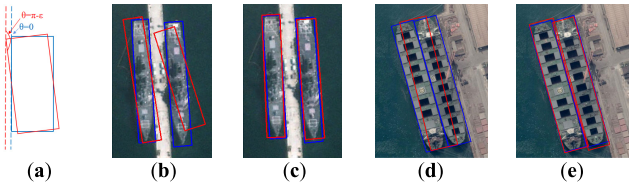

Fig. 12. (a) Loss mutation problem caused by angle periodicity. (b)–(e) Recognition results before and after introducing the IoU factor. Blue rectangles are ground truth, and red rectangles are predicted bounding boxes. (b) Inaccurate angle prediction near the angle of 0 without IoU factor. (c) Performance improvement after introducing the IoU factor. (d) "Riding box" problem without IoU factor. (e) Recognition results with the IoU factor.

no additional calculation expenses

$$\mathrm{CB}_{\mathrm{softmax}}(\mathbf{z}, y) = \frac{1-\beta}{1-\beta^{n_y}} \log\left(\frac{\exp(z_y)}{\sum_{j=1}^{C}\exp(z_j)}\right) \tag{11}$$

where β is a hyperparameter associated with the degree of class imbalance of the dataset, C is the total number of classes, and n_y is the number of samples in the ground-truth class.

For the rotated box regression task, we do not directly employ the smooth L1 function, such as in [17]–[19] and [29]–[31], but modify it with an IoU factor. The reason lies in the loss mutation problem caused by angle periodicity. As shown in Fig. 12(a), when the angle of the ground truth $\theta_{gt} = 0$ and the angle of the predicted box $\theta_2 = \pi - \varepsilon$ ($\varepsilon \to 0$), this prediction should be very successful because the two boxes look extremely closely. However, since the output of the smooth L1 function is positively correlated with $|\theta_{gt} - \theta_2|$, the loss will be terrifically high. Fig. 12(b) depicts the consequence, in which we can see that network is trying to avoid predicting the angle near 0 or π due to the unstable loss, which leads to an inaccurate angle prediction. We introduce the IoU factor [see (10)] to break the limitation of the smooth L1 function. In the modified-smooth L1 loss, |Factor_IOU| determines the magnitude of gradient descent, and $((L_{\mathrm{reg}}(v'_j, v_j))/(|L_{\mathrm{reg}}(v'_j, v_j)|))$ determines the gradient descent direction. When $\theta_2 = \pi - \varepsilon$, |Factor_IOU| ≈ 0. With modified-smooth L1 loss, as shown in Fig. 11(c), the network predicts more accurate angles. There are two items in the IoU factor: the first item measures IoU between the predicted box and the corresponding ground truth, telling the network that the predicted box and the corresponding ground truth should be as close as possible; the second item measures IoU between the predicted box and all other ground truth, telling the network that the predicted box and all other noncorresponding ground truth should be as far away as possible. It is natural to utilize the first term as the magnitude of gradient descent because the evaluation of network performance is always based on IoU between the predicted box and ground truth. The second item is introduced to solve the "riding box" problem in dense scenes [see Fig. 11(d)], which means a prediction box "riding" on two dense targets. Although the ground truth of rigid bodies in aerial view will not overlap theoretically, overlaps of ground-truth boxes are still inevitable in dense scenes since regular rectangles cannot entirely fit the contour of ships. Therefore, in the second term, we subtract IoU between ground truth corresponding to the predicted box and all other ground truths. Recognition results with the IoU factor are shown in Fig. 12(e), indicating that the modified-smooth L1 loss can effectively solve the "riding box" problem in dense scenes.

IV. EXPERIMENT

A. Datasets

1) Our Dataset for Oriented Ship Recognition: In view of the limitation of HRCS2016, such as missing annotation of small ships and incorrect annotation of ship categories, we build a new public available DOSR. The images in DOSR are collected from Google Earth. To expand the diversity of data, we collect images from quantities of countries, including the United States, China, France, Japan, and so on. The shooting time spans from 2001 to 2020. The detailed information of DOSR is listed in Table II; 1066 optical remote sensing images and 6127 ship instances are contained in DOSR. The image size is ranged from 600 to 1300 pixels with resolutions of 0.5–2.5 m. In DOSR, there are 832 nearshore scenes containing 5212 instances and 234 offshore scenes containing 915 instances. The dataset is divided into three sets: the training set with 523 images, the validation set with 223 images, and the testing set with 320 images. Fig. 13 displays examples of 20 fine-grained classes in DOSR. DOSR is a challenging dataset with the following properties.

1) *Fine-Grained Classes With Oriented Bounding Box:* Ships in DOSR are classified into 20 classes, and the instance amount of each class is shown in Table III. Each instance, no matter how small, is annotated with a five-parameter tuple (x, y, w, h, θ), which is consistent with parameters in oriented bounding box regression in Section III. As ship regulations vary from country to country, one challenge of DOSR is the huge intraclass distinction among some categories, such as cruise, auxiliary, and fishing boat. Unfortunately, the class imbalance problem remains unsolved in DOSR. Some rare classes, such as communication ships and submarines, are hard to collect, whereas some common ships, such as transports and yachts, are everywhere.
2) *Abundant Instances and Rich Scenes:* There is an average of 5.75 instances per image in DOSR, whereas this number in HRSC2016 is only 2.72, indicating that our DOSR provides more abundant instances and richer

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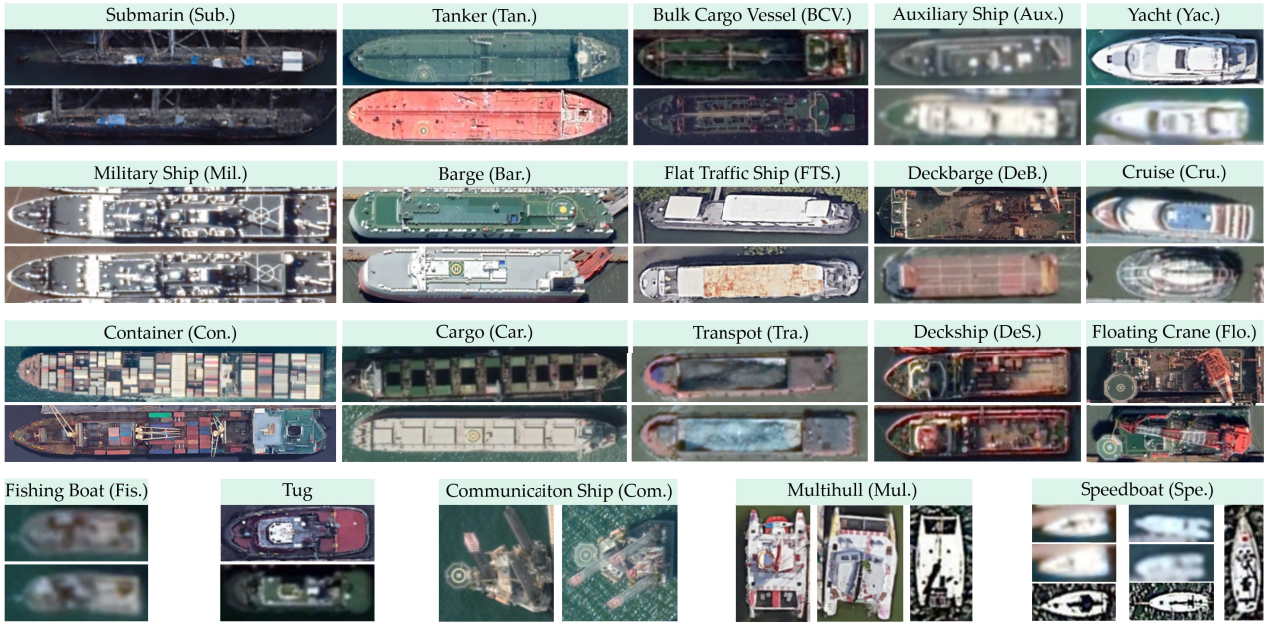

Fig. 13. Samples of 20 fine-grained ship classes in DOSR.

TABLE III
INSTANCE AMOUNT OF 20 CLASSES IN DOSR

Tra.	Yac.	Spe.	Aux.	Mli.	Tug	Fis.	BCV.	Car.	Con.
1682	1188	529	426	419	293	276	275	249	177
Cru.	**DeB.**	**Tan.**	**DeS.**	**FTS.**	**Flo.**	**Mul.**	**Bar.**	**Com.**	**Sub.**
165	100	86	69	59	38	36	34	14	12

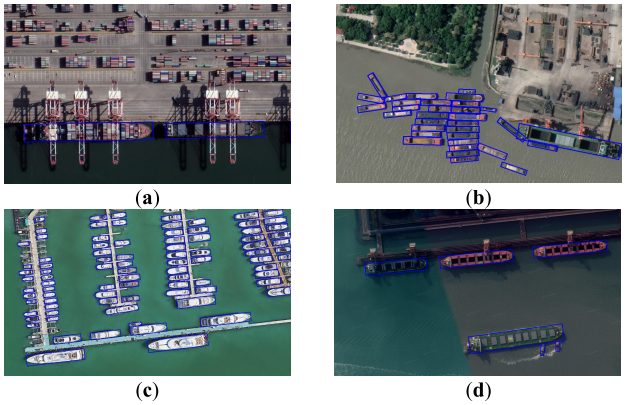

Fig. 14. Typical scenes in DOSR. (a) Clutter scenes. (b) Dense scenes. (c) Small scenes. (d) Large-scale difference scenes.

recognition scenes with denser ship distribution. Moreover, there are diverse scenes in DOSR, which is another challenge for recognition algorithms. Fig. 14 shows four typical scenes in DOSR, containing clutter scenes, dense scenes, small scenes, and huge scale variance scenes.

3) *Public Availability:* Our DOSR is publicly available. Readers can get DOSR on our Github: https://github.com/yaqihan-9898/DOSR.

2) HRSC2016: HRSC2016 is the most widely used dataset in previous ship recognition research. Thus, to validate our EIRNet fairly and comprehensively, we also include HRSC2016 into experiment datasets. Note that, although there are 29 fine-grained categories announced in HRSC2016, two categories have no training instance, one category has only one testing instance, and four categories belong to upper level categories. Therefore, there are only 22 valid categories in HRSC2016. We exclude Hovercraft, Yacht, and Cruise ship as did in [2]. Finally, only 19 categories are tested in experiments, containing *Submarine (Sub.), Nimitz (Nim.), Enterprise (Ent.), Arleigh Burke (Arl.), WhidbeyIsland (Whi.), Perry (Per.), Sanantonio (San.), Ticonderoga (Tic.), Admiral (Adm.), Austen (Aus.), Tarawa (Tar.), Container (Con.), Command ship A (Com.A), Car carrier A (Car.A), Container ship A (Con.A), Medical ship (Med.), Car carrier B (Car.B),* and *Midway (Mid).*

B. Experiment Setup

In this section, we introduce the experiment details, including experiment environment, detailed hyperparameter settings, and evaluation metrics for network performance.

1) *Experiment Environment:* We implement the proposed network in the Tensorflow framework and conduct all experiments on Nvidia GTX 1080Ti GPU with 11-GB memory.

2) *Experiment Settings:* ResNet101 is chosen to be the backbone and initialized with the pretrained model on ImageNet. We randomly flipped the images horizontally and vertically to augment data. Parameters in C_1 block and C_2 block are frozen to accelerate convergence. The weight decay is set to 0.0001, and the momentum is 0.9.

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TABLE IV
ABLATION STUDIES OF EACH COMPONENT IN EIRNET ON DOSR

Net name	Net component	Modified-Smooth L1	Upper-level	Mask-RPN	BBox	BCV.	Fis.	DeB.	Yac.	FTS.	Mul.	Tug	Com.	Spe.	Car.	Cru.	Flo.	Tan.	DeS.	Sub.	Con.	Bar.	Tra.	Aux.	Mil.	mAP
Net1	Baseline				R	50.68	19.75	20.89	45.14	20.01	43.8	58.11	50.00	27.37	81.98	49.5	16.23	64.42	16.67	9.19	66.47	24.76	66.66	26.46	30.69	39.44
Net2	Baseline				H+R	52.98	10.07	20.57	47.10	48.95	36.15	56.87	55.48	33.81	80.64	47.41	20.58	82.58	24.03	0.17	56.82	35.06	76.43	19.21	33.98	41.94
Net3	Baseline+DFF				H+R	61.81	24.64	28.82	51.66	39.42	37.76	62.35	47.24	35.98	81.36	53.15	4.80	68.86	23.37	30.03	58.54	40.11	76.47	32.08	43.33	45.09
Net4	Baseline+DFF+DMAM				H+R	66.86	37.97	44.72	59.75	37.86	52.87	65.00	40.00	38.31	86.79	55.69	33.49	78.24	40.38	62.94	71.92	49.76	76.98	40.06	47.93	54.38
Net5	Baseline+DFF+DMAM	√			H+R	67.02	56.66	25.88	70.48	41.63	59.09	74.50	14.29	38.69	86.69	71.35	47.39	78.28	29.49	22.45	79.74	64.58	76.63	52.39	51.88	55.45
Net6	Baseline+DFF+DMAM	√	√		H+R	65.05	63.67	47.22	64.33	54.73	54.42	73.20	41.68	47.68	86.42	68.25	40.58	60.48	23.86	84.85	73.29	37.10	77.82	45.10	49.04	57.94
Net7 (ours)	Baseline+DFF+DMAM	√	√	√	H+R	67.52	55.65	48.59	68.53	70.48	57.36	75.52	58.26	42.53	87.87	67	55.3	74.9	56.65	59.6	78.63	30.46	78.34	45.98	48.59	**61.39**

The learning rate is initialized to 0.001. For DOSR, we train the model 75k iterations totally and decrease the learning rate to 0.0001 and 0.00001 at 30k and 60k iterations, respectively. For HRSC2016, the model is trained 60k iterations, and the learning rate decays ten times after 25k iterations and 100 times after 45k iterations. Mask weights α_1 and α_2 in (7) are 0.5. For hyperparameters in the loss function, α in IoU factor [see (10)] is 1.0, and β in the CB softmax cross-entropy loss is set to 0.97 for DOSR and 0.90 for HRSC2016. The multitask tradeoff weights $\lambda_{1,2,3}$ in (9) are set to 1.0 and $\lambda_{4,5}$ are set to 2.0. As a little trick, λ_4 for fine-grained classification branch is doubled after 30k iterations to make the network pay more attention to classification tasks in the late stage of training. The minibatch per iteration involves one anchor batch and 256 RoIs. In preprocessing, the shorter edge of each image is resized to 600 pixels, and the max edge is limited to 1000 pixels. R-NMS threshold for postprocessing is 0.15.

3) *Evaluation Metrics:* We choose average precision (AP) to evaluate the recognition performance on each class and use mean AP (mAP) to assess the overall performance. The confidence IoU threshold for AP calculation is 0.5.

C. Ablation Studies

In this section, we discuss the effect of individual components in our EIRNet and the influence of some detailed network settings. We adjust Faster RCNN with FPN to oriented regression task and choose it as our baseline. Tables IV and V show the overall experiment results of ablation studies on DOSR and HRSC2016, respectively. Fig. 15 displays recognition results of seven typical scenes in DOSR with different networks in ablation studies. Samples of recognition results on HRSC2016 are illustrated in Fig. 16.

1) Effect of DFF-Net: DFF-Net can efficiently fuse features from all backbone layers in both up-down and bottom-up directions, and the introduction of channel attention in DFF-Net reduces redundancy of excessive information. As shown in Tables IV and V, the baseline with FPN (Net2) only gets mAP 41.94% and 72.32% on DOSR and HRSC2016, respectively. After replacing FPN with DFF-Net in baseline (Net3), mAP raises by 3.15% and 2.63%. The performance improvement most attributes to the AP increase for some small-scale ship categories, such as Fis. (+14.57%), Aux. (+12.87%), and DeB. (+8.25%). In order to further explore the superiority of DFF-Net and the effectiveness of each design concept, we remove or substitute some components in DFF-Net and compare the performance of these variant networks. The experiment results are listed in Table VI. There are three variant networks in total. DFFv1 only changes the fusion method in FPN to the dense fusion method, which means each layer fused with all layers above it. Based on DFFv1, DFFv2 introduces a bottom-up fusion route. In other words, DFFv2 is a variant version of the proposed network that removes the channel attention module. DFFv3 shares the same structure with the proposed DFF-Net, except that it replaces the channel attention module from CBAM to the popular SE-Net [63]. To further explore the effect of DFF-Net on different target scales, we measure mAP on small ships, middle ships, and large ships. As there are no small ships and few middle ships in HRSC2016, we only implement experiments on DOSR. Table VI displays the results. The proposed DFF-Net achieves the highest mAP, which attributes to increased accuracy at all scales, especially small ships. It is worth noting that DFFv1 performs worse than FPN. This is because the dense fusion method brings severe information redundancy. Besides, the good performance of DFFv2 demonstrates the efficiency of the introduction of the bottom-up fusion direction. The channel attention mechanism further assists DFF-Net to achieve the best results, and experiment results show that the channel attention module in CBAM is more suitable for our EIRNet than SE-Net.

2) Effect of DMAM: DMAM can be freely embedded in the network to filter out clutter and highlight targets, especially dense targets. Comparing experiment results of Net3 and Net4 in Tables IV and V, we can find that DMAM boosts the performance on DOSR from 45.09% to 54.38% and increases mAP greatly on HRSC2016 from 74.95% to 78.65%. The reason for different effects of DMAM on different datasets lies in that DOSR has much more dense and complex scenes and can more prominently reflect the excellent ability of DMAM. We also study the role of the shortcut in DMAM. The performance of EIRNet before and after removing the shortcut is shown in Table VII. Without the shortcut, EIRNet only gets mAP of 56.09% and 76.78%, demonstrating the essential role of shortcuts. The reason is that the response

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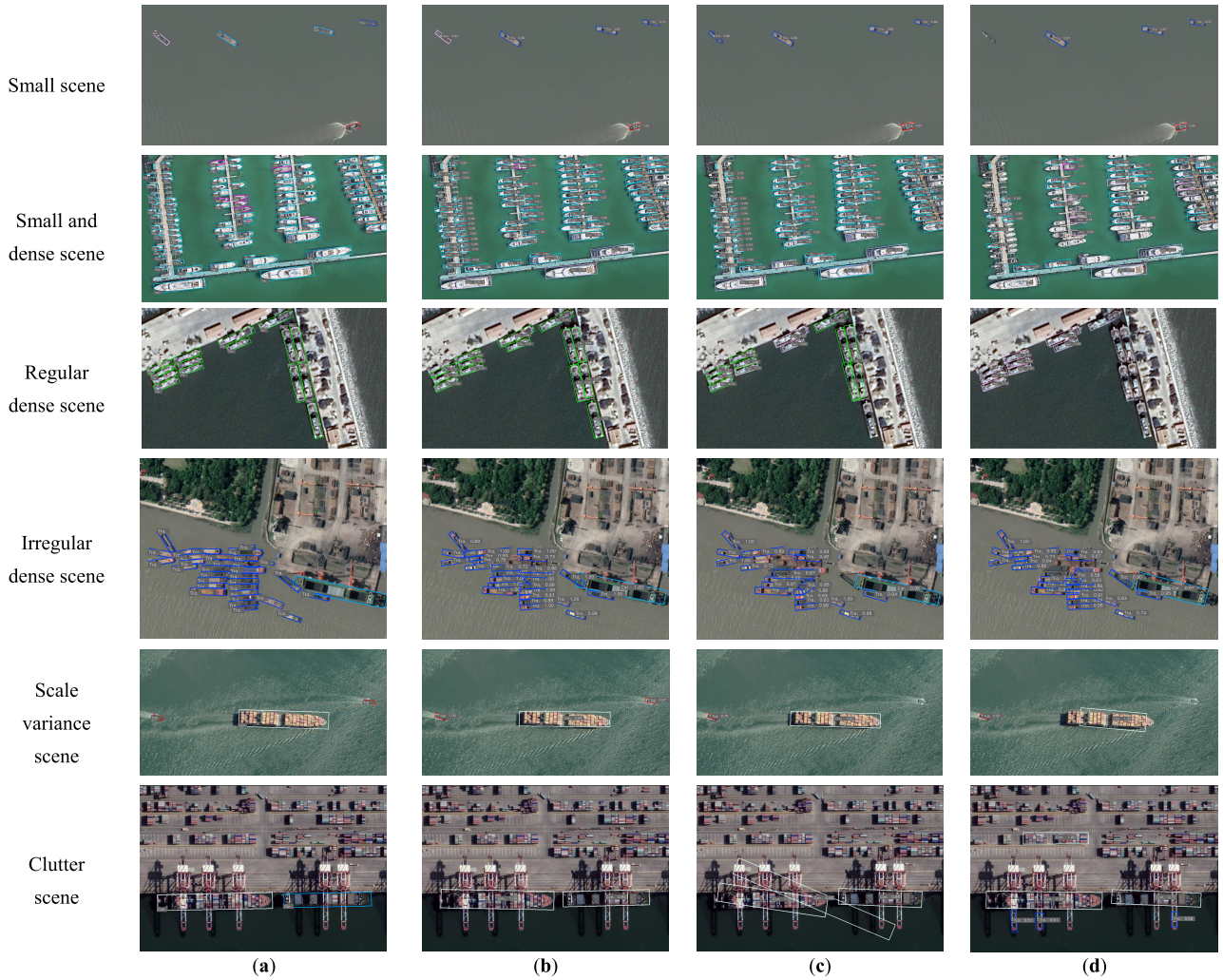

Fig. 15. Performance of three networks in ablation studies in seven typical scenes of DOSR. (a) Ground truth. (b) Proposed EIRNet (Net7). (c) Baseline + DMAM + DFF (Net4). (d) Baseline with H + R bbox (Net2). The comparison between (c) and (d) proves that DMAM + DFF can boost the ability to recognize small targets and dense targets and suppress clutter. Results in (b) demonstrate that modified-smooth L1 + upper level + Mask-RPN can further improve classification ability and recognition performance in dense scenes. Samples of 20 fine-grained ship classes in DOSR.

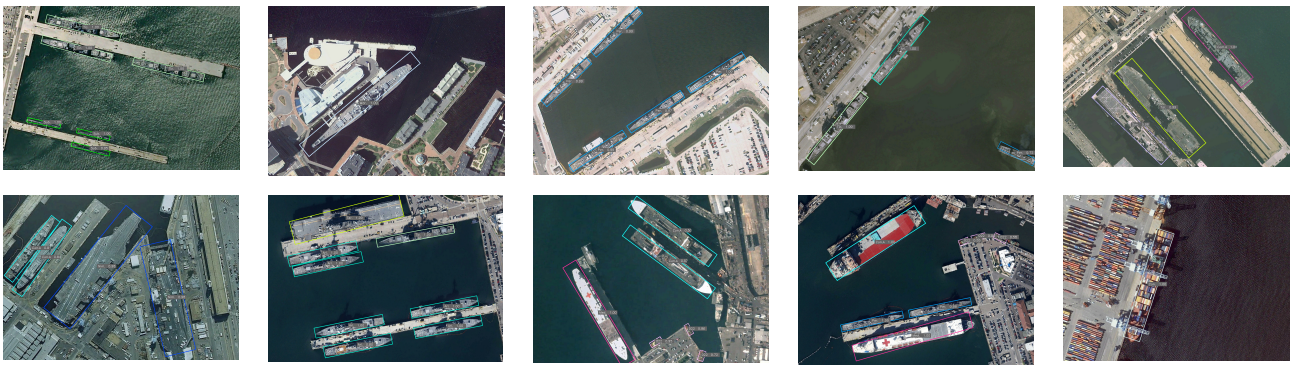

Fig. 16. Samples of recognition results on HRSC2016.

of large background areas will become 0 if there is no shortcut, resulting in quantities of nodes in subsequent network responses to 0 and further leading to network degradation.

In addition, we conduct a group of experiments to figure out the influence of the weight hyperparameters α_1 and α_2 in (7), and the results are listed in Table VII. When using a single attention mask (either α_1 or α_2 equals 0), the network's performance is lower than that in the dual-mask situation, indicating the effectiveness of the dual-mask mechanism. To further compare the contribution of masks M_1 and M_2 to the network, we also set two experiments in the case of $\alpha_1 = 1, \alpha_2 = 2$ and $\alpha_1 = 2, \alpha_2 = 1$. The results imply that M_1 plays a more significant role in HRSC2016, whereas M_2 contributes more in DOSR. Therefore, it is natural to

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TABLE V
ABLATION STUDIES OF EACH COMPONENT IN EIRNET ON HRSC2016

Net name	Net component	Modified-Smooth L1	Upper-level	Mask-RPN	BBox	Sub.	Nim.	Ent.	Arl.	Whi.	Per.	San.	Tic.	Adm.	Aus.	Tar.	Con.	Com.	Car.A	Con.A	War.	Med.	Car.B	Inv.	mAP
Net1	Baseline				R	49.49	75.05	43.01	78.16	65.99	72.87	69.8	79.39	54.55	60.44	86.50	70.53	71.32	74.78	67.08	64.36	94.85	88.02	61.31	69.87
Net2	Baseline				H+R	51.89	85.17	63.45	83.00	57.52	80.57	70.24	82.95	84.85	46.89	93.37	71.13	63.56	73.25	69.99	55.34	96.06	68.51	76.29	72.32
Net3	Baseline+DFF				H+R	53.38	75.62	68.44	83.68	61.55	74.12	79.09	79.19	100.00	50.09	88.88	62.08	73.42	87.10	73.56	64.70	98.48	72.43	78.33	74.95
Net4	Baseline+DFF+DMAM				H+R	59.02	83.93	53.00	85.43	74.31	85.30	97.01	88.73	100.00	79.45	89.53	51.25	83.17	92.54	66.64	40.91	96.69	93.20	74.38	78.65
Net5	Baseline+DFF+DMAM	√			H+R	48.73	82.77	60.39	87.48	75.24	86.71	94.10	86.42	100.00	74.42	87.81	60.66	82.09	80.52	68.79	72.73	99.17	86.89	80.66	79.77
Net6	Baseline+DFF+DMAM	√	√		H+R	56.95	83.18	71.87	86.59	76.57	85.46	81.93	86.01	100.00	79.90	87.81	64.94	75.88	78.98	63.22	72.73	99.17	94.17	80.68	80.32
Net7 (ours)	Baseline+DFF+DMAM	√	√	√	H+R	74.38	89.54	51.25	88.94	77.97	82.49	93.78	90.22	100.00	86.99	90.91	65.24	88.65	88.14	69.35	73.47	95.87	92.35	93.39	**83.84**

TABLE VI
EFFECT OF EACH COMPONENT OF DFF-NET ON DOSR

Net	Channel attention	Bottom-up	Top-down	AP_S	AP_M	AP_L	mAP
FPN	--	--	Normal	40.07	56.83	63.77	58.74
DFFv1	--	--	Dense	45.33	54.31	62.98	57.19
DFFv2	--	√	Dense	40.63	57.99	65.07	60.40
DFFv3	SE	√	Dense	39.85	49.87	59.62	57.48
proposed DFF-Net	CBAM	√	Dense	**45.85**	**58.71**	**67.22**	**61.39**

TABLE VII
EFFECT OF DIFFERENT SETTINGS OF DMAM ON DOSR AND HRSC2016

Shortcut	α_1	α_2	Mask value range	DOSR	HRSC2016
--	0.5	0.5	[0,1]	56.09	76.78
√	1	0	[0,1]	57.91	80.63
√	0	1	[0,1]	58.73	80.04
√	1	1	[0,2]	56.16	79.10
√	2	1	[0,3]	57.45	80.84
√	1	2	[0,3]	59.28	80.32
√	2	2	[0,4]	57.56	78.60
√	0.25	0.25	[0,0.5]	59.80	82.35
√	0.5	0.5	[0,1]	**61.39**	**83.84**

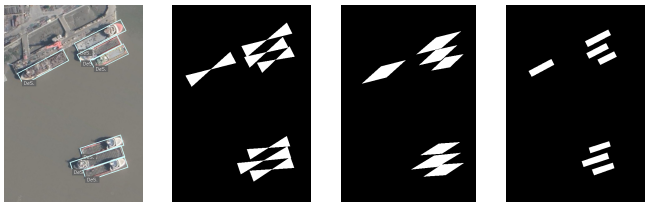

Fig. 17. Shape designs of Mask-2. Apart from the proposed shape, we test the other two shapes of Mask-2 to solve the ship distinguish problem in dense scenes. Test Shape 1 is the largest inscribed rhombus of ground truth. Test Shape 2 is a rectangle with half-length and width of ground truth.

make a tradeoff, that is, making the two hyperparameters equal. The last three experiments in Table VII demonstrate that the network performs best when α_1 and α_2 are set to a moderate value of 0.5. This is because the excessively high value of mask will dilute the role of shortcut and cause network degradation, and the low value of mask will obviously weaken the effect of DMAM.

To verify the effectiveness of the shape design of Mask-2, we also test the other two mask shapes shown in Fig. 17.

TABLE VIII
INFLUENCE OF DIFFERENT ANCHOR TYPES AND ANCHOR ASPECT RATIOS

Anchor Type	Anchor Aspect Ratios	DOSR		HRSC2016	
		mAP	Speed/s	mAP	Speed/s
Rotated	$\{i, \frac{1}{i}\}(i = 1,2,3,4,5,6,7)$	61.05	0.4215	84.56	0.3413
Horizonal	$\{i, \frac{1}{i}\}(i = 1,2)$	59.29	0.2548	82.96	0.1798
Horizonal	$\{i, \frac{1}{i}\}(i = 1,2,3)$	60.03	0.2636	82.01	0.1957
Horizonal	$\{i, \frac{1}{i}\}(i = 1,2,3,4)$	61.39	0.2708	83.84	0.2031
Horizonal	$\{i, \frac{1}{i}\}(i = 1,2,3,4,5)$	58.93	0.2901	83.57	0.2168

Shape 1 and Shape 2 only get mAP of 59.25% and 57.63% on DOSR, respectively.

3) Effect of Mask-RPN: The core idea of Mask-RPN is reusing mask information from DMAM to assist traditional RPN, which is an effective approach to improve performance with little additional computing expense. From Net6 and Net 7 in Tables IV and V, we can see that Mask-RPN improves mAP by 3.45% (from 57.94% to 61.39%) and 3.52% (from 80.10% to 82.96%) on DOSR and HRSC2016, respectively.

In addition, comparison experiments are conducted to evaluate the performance of rotated anchors and horizontal anchors. The average aspect ratio of horizontal and oriented ground truth in DOSR is 2.38 and 4.38, respectively. Thus, we set four experiments for horizontal anchors with different anchor aspect ratios around 2.38 and adopt rotated anchor settings in [29]. As shown in Table VIII, horizontal anchors with aspect ratios of $\{i, (1/i)\}(i = 1, 2, 3, 4)$ perform best. Rotated anchors cost much more time than horizontal anchors. However, performance improvement does not match the computation expense. What is worse, rotated anchors reduce mAP on DOSR, which may be owning to dense scenes in the DOSR affect the accuracy of proposal generation and RROI pooling.

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TABLE IX
UPPER LEVEL CLASS SETTINGS FOR DOSR

L1	Ship																			
L2	Merchant ship					Passenger ship						Work ship							Military ship	
L3	BCV.	Car.	Tan.	Con.	Bar.	Tra.	Yac.	FTS.	Mul.	Spe.	Cru.	Fis.	DeB.	Tug	Com.	Flo.	DeS.	Aux.	Sub.	Mil.

TABLE X
UPPER LEVEL CLASS SETTINGS FOR HRSC2016

L1	Ship																		
L2	Aircraft carrier					warcraft									Merchant ship				Submarine
L3	Nim.	Ent.	Adm.	Tar.	Inv.	Arl.	Whi.	Per.	San.	Tic.	Aus.	Com.	War.	Med.	Con.	Car.A	Con.A	Car.B	Sub.

4) Effect of Upper Level Classification Branch and Horizontal Bounding Box Regression Branch: In contrast with baseline, there are two new prediction branches in EIRNet: oriented bounding box regression branch and upper level classification branch. Although the oriented bounding box regression branch can meet the needs of predicting rotated boxes, we remain the horizontal bounding box regression branch in EIRNet. The experiment results in Tables IV and V prove the necessity of predicting horizontal boxes. Baseline without horizontal bounding box regression branch (Net1) gets mAP of 39.44% and 69.87%, whereas mAP reaches 41.94% and 72.32% when horizontal boxes and rotated boxes are predicted simultaneously (Net2). In the training process, the horizontal bounding box regression branch assists EIRNet to locate objects easier and makes the network pay more attention to learn overly sensitive angle information.

As mentioned in Section III, the upper level classification branch is designed to help the network learn the interclass relationships and further improve classification capability under the class-imbalance problem. Detailed settings of the upper level class of DOSR and HRSC2016 are shown in Tables IX and X. The experiments results show that, after introducing upper level classification branch (Net6), the performance of the network increases from 55.45% to 57.94% on DOSR and from 79.77% to 80.32% for HRSC2016. The proposed EIRNet only employs the bottom-level L3 class and its upper level L2 class to complete classification tasks. We also experiment with the performance after introducing the top-level L1 class (ship or not ship). As shown in Table XI, the L1+L2+L3 mode only gets mAP of 56.96% and 80.97%, implying a large class span between L1 and L3, and the lack of category information in L1 may mislead the learning focus. Therefore, we choose the L2+L3 mode to realize a more accurate classification.

5) Effect of Modified-Smooth L1 Loss: To avoid loss mutation problems caused by angle periodicity, we modify smooth L1 with an IoU factor in (10). Results of Net4 and Net5 in Tables IV and V prove the effect of the modified-smooth L1 loss. The modified-smooth L1 loss improves mAP by 1.07% and 1.12% on DOSR and HRSC2016, respectively.

The loss curve before and after modifying smooth L1 is illustrated in Fig. 18. It can be observed that, although the overall downward trend of the loss curves under the two loss functions is similar, the entire training process is extremely unstable with traditional smooth L1 due to some abnormally high losses, whereas the loss curve with modified-smooth L1 loss is obviously much more stable.

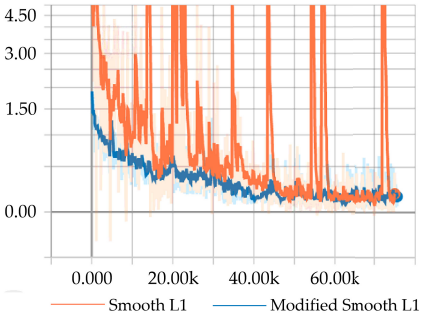

Fig. 18. Loss curve of rotated box regression branch with smooth L1 loss and modified-smooth L1 loss. The curves are obtained from Tensorboard with a smoothing parameter of 0.890. Orange curves are smooth L1 loss, and blue curves are a modified-smooth L1 loss. The lighter curves are original loss values, and the darker curves are smoothed loss values.

TABLE XI
EFFECT OF DIFFERENT COMBINATIONS OF CLASSIFICATION BRANCHES ON DOSR AND HRSC2016

Classification branches	DOSR	HRSC2016
L3+L2+L1	56.96	80.97
L3+L2(proposed)	**61.39**	**83.94**

6) Influence of Different Network Heads: We explore the influence of three different network heads on ship recognition task, containing: 1) two traditional 4096 channel full connection layers with two dropout layers (drop rate = 0.5); 2) light-head proposed in [68]; and 3) C5 block of backbone that we used. All three heads share the same experiment settings, such as learning rate and training iteration. The experiment results are displayed in Table XII. C5 block performs best, whereas the other two network heads get unexpectedly poor results, especially for two full connection layers. We also tried to adjust experiment settings for 2fc and light-head to exert their best ability. However, such adjustment does not bring much improvement. Therefore, C5 block is a better choice for EIRNet.

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TABLE XII
INFLUENCE OF DIFFERENT NETWORK HEADS ON DOSR AND HRSC2016

Head	DOSR	HRSC2016
2FC	27.89	52.98
Light-head	46.23	76.33
C5 (ours)	**61.39**	**83.94**

D. Comparison With the State of the Art

In this section, we compare the performance of the proposed EIRNet with nine state-of-the-art recognition approaches, including Faster RCNN with FPN and oriented bounding box regression branch (FR-FPN-O), R2CNN [30], RRPN [29], SCRDet [45], RetinaNet-O [69], R3Det [70], RSDet [71], SCRDet++ [72], and ReDet [35]. Among them, FR-FPN-O, R2CNN, RRPN, and RetinaNet-O are general-oriented object recognition networks, and the rest five approaches are all designed for remote sensing object recognition. In more detail, FR-FPN-O, R2CNN, RRPN, SCRDet, and ReDet adopt the two-stage strategy, and the single-stage networks involve RetinaNet-O, R3Det, RSDet, and SCRDet++. For a comprehensive evaluation, all methods are implemented on both DOSR and HRSC2016. To be fair, except for ReDet, which develops a new backbone, we implement all networks above with RestNet101. We adjust the NMS threshold to 0.15 to exert their best performance on the fine-grained ship recognition task. Data augmentation methods and other hyperparameter settings are consistent with the original papers. The performance (mAP) of these models on DOSR and HRSC2016 is listed in Tables XIII and XIV, respectively. We observe that two-stage networks do have an advantage in accuracy, and one-stage networks are faster. It is obvious that our EIRNet achieves state-of-the-art performance both on DOSR and HRSC2016. The speed of EIRNet is also comparable among two-stage networks.

E. Discussion

Although our EIRNet gets state-of-the-art results on DOSR and HRSC2016, the performance still cannot meet the requirement of practical applications. The remaining unsolved problems can be analyzed from two perspectives: 1) robustness of detection for all ship scales and various complex scenes and 2) accuracy of fine-grained ship classification.

For the robustness of ship detection, the sensitivity of the NMS threshold is a significant issue. As we analyzed in Section I, a high aspect ratio is a distinguishing characteristic of ships. However, in the case of a high aspect ratio, a small deviation between the predicted box and ground-truth box will cause a rapid decrease in IoU, making the network performance susceptible to the NMS threshold. Fig. 19 illustrates the mAP of EIRNet under different NMS thresholds. DOSR and HRSC2016 get their highest mAP at the threshold of 0.23 and 0.09, respectively, indicating that the NMS threshold is sensitive to datasets. We adopt the NMS threshold of 0.15 in ablation studies as its performance is relatively stable under different datasets and experimental settings. In addition, DOSR, which contains more dense scenes, is much more

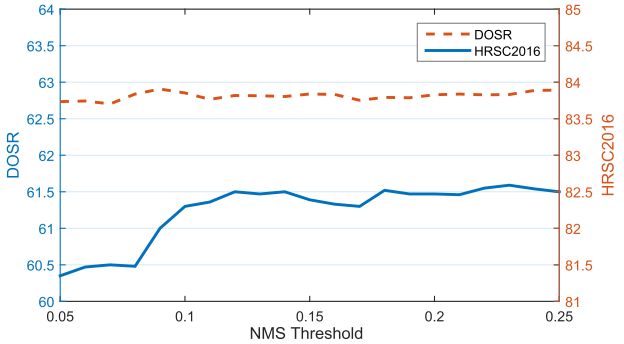

Fig. 19. Performance of EIRNet on DOSR and HRSC2016 under different NMS thresholds. The blue curve is mAP on DOSR, and the green curve is mAP on HRSC2016.

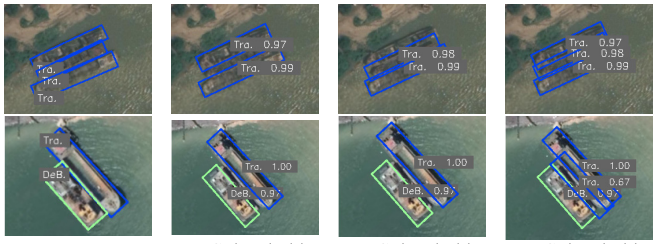

Fig. 20. Detection results with different NMS thresholds.

sensitive to the NMS threshold than HRSC2016. To figure out the impact of the NMS threshold more intuitively, we display two patches of detection results with different thresholds in Fig. 20, which indicates that a low threshold will mistakenly suppress the target, whereas a high threshold may cause false alarms. Therefore, it is really hard to select a proper threshold. Even if we find the best threshold for a certain dataset through a lot of experiments, the threshold may be not suitable for all scenes. We also tried other NMS models, e.g., soft-NMS [73]. Unfortunately, we got disappointing results: soft-NMS causes many false alarms and only gets mAP of 57.53% (−3.67%) on DOSR. Current improved NMS algorithms represented by soft-NMS are mainly designed for eliminating the influence of occlusion between targets in a horizontal perspective. Nevertheless, in a bird's eye view, there is no occlusion between objects at all. Therefore, such NMS algorithms fail to match the needs of recognition tasks in remote sensing images. In our future work, a new NMS algorithm will be explored to smash the dilemma.

The accuracy of ship classification is mainly limited by: 1) interclass similarity and intraclass variance and 2) class imbalance. This view can be evidenced by the confusion matrix shown in Fig. 21. We find classes with huge intraclass variance, such as Cru., Aux., and Mli., which are easier to confuse the network and be classified to other classes. In addition, we notice that misclassification occurs in some class groups with interclass similarity. For example, the network often fails to distinguish Flo., Com., and DeS. The reason may be the interclass similarity of this group. More specifically, most instances of Flo. and DeS. are both square and similar in color, and Com. and Flo. both have spire-type devices. A similar mistake also happened in the group of Yac. and Spe.

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TABLE XIII
COMPARISON OF PERFORMANCE (AP) ON DOSR

Method	BCV.	Fis.	DeB.	Yac.	FTS.	Mul.	Tug	Com.	Spe.	Car.	Cru.	Flo.	Tan.	DeS.	Sub.	Con.	Bar.	Tra.	Aux.	Mil.	mAP	Speed(s)
FR-FPN-O	37.14	8.79	7.98	47.30	25.59	48.20	48.11	50.80	28.54	80.57	49.81	8.64	65.4	15.54	10.03	56.69	12.21	69.35	10.78	32.18	35.68	0.2033
R2CNN	56.47	36.86	38.84	57.14	26.19	54.55	58.98	32.03	39.27	76.7	52.99	5.85	50.87	42.41	13.64	75.57	46.08	66.85	43.54	36.85	45.58	0.3513
RRPN	62.89	42.80	33.15	47.15	43.66	52.50	74.57	27.27	31.19	83.30	57.52	33.10	64.89	20.55	54.55	78.61	52.95	78.26	36.06	36.63	50.58	0.3703
SCRDet	65.21	54.26	44.02	53.33	49.46	38.31	66.67	40.74	32.19	86.66	61.50	21.95	87.68	32.84	54.55	80.58	49.91	73.68	42.78	49.46	54.29	0.2602
RetinaNet-O	49.52	26.24	36.86	50.63	27.10	35.00	70.45	49.74	33.68	68.70	35.87	11.93	38.11	2.04	0.00	55.99	12.16	66.27	29.66	34.06	36.70	0.2686
R3Det	65.60	44.78	28.49	64.21	49.14	53.03	68.16	38.83	35.97	84.24	63.36	34.35	71.63	26.70	57.61	74.77	42.70	74.03	27.19	48.48	52.66	0.2043
SCRDet++	61.22	44.45	36.06	67.58	62.80	61.62	81.09	66.09	63.51	76.28	66.14	13.91	77.01	58.98	32.73	70.00	18.75	76.82	47.27	44.22	56.33	0.3531
RSDet	55.27	10.76	22.84	59.78	52.56	47.30	58.78	63.64	57.22	75.32	39.60	16.94	33.13	14.23	45.69	68.82	26.19	75.30	33.96	45.64	45.15	0.3612
ReDet	64.28	43.17	27.87	75.65	65.06	41.59	80.10	37.67	67.95	85.06	62.40	37.54	83.44	36.58	34.85	76.64	44.31	86.03	38.64	57.48	57.32	**0.0833**
EIRNet(ours)	67.52	55.65	48.59	68.53	70.48	57.36	75.52	58.26	42.53	87.87	67.00	55.30	74.90	56.65	59.60	78.63	30.46	78.34	45.98	48.59	**61.39**	0.2708

TABLE XIV
COMPARISON OF PERFORMANCE (AP) ON HRSC2016

Method	Sub.	Nim.	Ent.	Arl.	Whi.	Per.	San.	Tic.	Adm.	Aus.	Tar.	Con.	Com.	Car.A	Con.A	War.	Med.	Car.B	Inv.	mAP	Speed(s)
FR-FPN-O	55.46	71.13	59.64	80.74	60.81	71.08	79.76	79.10	100.00	56.63	77.05	58.36	70.13	75.71	63.33	55.88	92.87	86.88	42.23	70.36	0.1636
R2CNN	33.93	60.05	27.70	77.11	53.93	60.82	78.99	77.81	100.00	75.11	77.82	29.72	64.13	76.78	53.30	91.71	63.77	84.70	45.45	64.89	0.3081
RRPN	55.33	73.47	53.54	88.42	67.08	75.20	86.98	86.22	27.27	75.78	98.61	52.21	80.07	81.21	71.44	54.55	92.12	82.95	53.60	71.37	0.3840
SCRDet	51.23	74.12	65.71	79.57	57.89	75.41	52.72	81.08	84.85	55.86	89.72	60.87	67.47	78.83	60.91	46.59	91.49	80.26	95.75	71.07	0.1812
RetinaNet-O	44.91	68.66	51.94	79.96	49.65	71.52	27.93	84.52	100.00	45.97	85.21	52.42	42.95	74.04	42.20	13.64	80.48	53.67	76.03	60.30	0.1451
R3Det	76.47	72.52	49.56	84.93	53.49	83.57	67.32	87.08	100.00	66.42	88.51	52.05	55.06	90.91	59.39	74.26	85.04	69.39	44.40	71.60	0.1884
SCRDet++	57.32	77.39	49.94	86.31	77.41	81.36	83.76	85.49	84.85	82.65	86.14	59.70	68.17	85.18	51.05	72.73	85.08	87.70	90.91	76.48	0.1939
RSDet	55.67	68.09	44.10	74.61	60.88	80.54	43.46	87.19	100.00	59.63	76.58	41.16	49.24	77.38	44.92	64.42	61.13	49.86	79.22	64.11	0.1866
ReDet	88.19	81.77	64.78	88.09	72.62	88.00	86.37	90.55	100.00	75.88	99.65	77.77	66.17	86.63	84.54	71.52	92.42	83.26	72.37	82.67	**0.0730**
EIRNet(ours)	74.38	89.54	51.25	88.94	77.97	82.49	93.78	90.22	100.00	86.99	90.91	65.24	88.65	88.14	69.35	73.47	95.87	92.35	93.39	**83.84**	0.2031

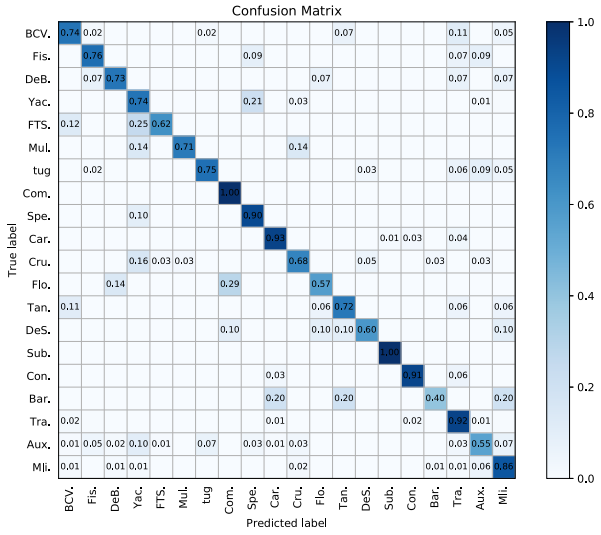

Fig. 21. Confusion matrix on DOSR.

and the group of Tan. and BCV. Another significant problem is a class imbalance, which inevitably leads to overfitting. Although tricks such as data augmentation and CB softmax cross-entropy loss are adopted to fight the negative impact of class imbalance, the risk of overfitting remains high. For example, five classes with the worst results in Fig. 21 are Bar., Aux., Flo., FTS., and Cru. Three of them contain instances less than 60. Therefore, we will further study effective algorithms to break the limitation analyzed above in our future work and enrich images in DOSR to solve the class imbalance problem once and for all.

V. CONCLUSION

In this article, we proposed an EIRNet for oriented fine-grained ship recognition tasks in remote sensing images. A DFF-Net was designed to cope with huge scale variations by efficiently and fully reusing the features from all backbone layers with both bottom-up and top-down fusion directions. Besides, we proposed a pixel-level attention network with a DMAM to improve the accuracy and robustness in dense and complex scenes. The mask generated by DMAM was further employed in Mask-RPN to improve the efficiency of proposal generation. To implement the prediction of oriented bounding boxes, we introduced a five-parameter branch and modified the regression loss. Moreover, we developed an upper level classification branch to guide the learning of interclass relationships, which further improved the performance. Another important contribution of this article was the establishment of a new public 20-class dataset called DOSR for fine-grained ship recognition. As far as we knew, our EIRNet achieved state-of-the-art performance on DOSR and another widely used dataset HRSC2016.

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In future work, we will enrich the ship instances in DOSR to diminish class imbalance from the root. Meanwhile, we will also study efficient approaches to reduce the risk of overfitting in the classification task. In addition, NMS algorithms will be further explored to address the inaccurate selection of final boxes in dense scenes.

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Yaqi Han received the B.S. degree from the School of Information and Communication Engineering, University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 2020, where she is currently pursuing the M.E. degree.

Her research interests include image processing, computer vision, and object detection and recognition.

Xinyi Yang received the B.S. degree from the School of Optoelectronic Science and Engineering, University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 2019, where she is currently pursuing the M.E. degree with the School of Information and Communication Engineering.

Her research interests include image processing, computer vision, object detection, and point cloud processing.

Tian Pu received the Ph.D. degree from the Beijing Institute of Technology, Beijing, China, in 2002.

He is currently a Lecturer with the School of Information and Communication Engineering, University of Electronic Science and Technology of China (UESTC), Chengdu, China. His research interests include image processing, computer vision, and medical image analysis.

Zhenming Peng (Member, IEEE) received the Ph.D. degree in geodetection and information technology from the Chengdu University of Technology, Chengdu, China, in 2001.

From 2001 to 2003, he was a Post-Doctoral Researcher with the Institute of Optics and Electronics, Chinese Academy of Sciences, Chengdu. He is currently a Professor with the University of Electronic Science and Technology of China, Chengdu. His research interests include image processing, signal processing, and target recognition and tracking.

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